25901391 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMAT202 - OPERATIONS RESEARCH

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. When is a solution said to be infeasible? [A] / [CO2]

2. What are objective function coefficients? Give an example. [K] / [CO1]

3. Define unbounded solution of a linear programming problem. [K] / [CO3]

4. How is Zj calculated in the simplex method? [U] / [CO3]

5. What is unbalanced transportation problem? [K] / [CO5]

6. Differentiate between balanced and unbalanced transportation problem [K] / [CO5]

7. Find row reduced cost table for the following assignment problem. [U] / [CO6]

	1	2	3	4
1	15	13	14	17
2	11	12	15	13
3	13	12	10	11
4	15	17	14	16

8. How will you handle the following situations in assignment problem [U] / [CO6]
1) Maximization 2) Unbalanced problem

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Explain how a general linear programming problem can be converted into standard form with an example. [U] / [CO1]

10. Solve graphically: [A] / [CO2]
Maximize $Z = 2x_1 + 4x_2$
Subject to $x_1 + x_2 \leq 10, x_1 \leq 6, x_2 \leq 8, x_1, x_2 \geq 0.$

11. Consider the linear programming problem: [A] / [CO3]

$$\begin{aligned} \text{Maximize} \quad & Z = 3x_1 - x_2 \\ \text{subject to,} \quad & 2x_1 + x_2 \leq 2 \\ & x_1 + 3x_2 \leq 1 \\ & x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Find the initial basic feasible solution. Does the problem have degenerate solution? Give reason.

12. Obtain the dual of the following primal problem $Minimize\ Z = 3x_1 - 2x_2 + x_3$ subject to the constraints [A] / [CO4]

$2x_1 - 3x_2 + x_3 \leq 5$
$4x_1 - 2x_2 \geq 9$
$-8x_1 + 4x_2 + 3x_3 = 8, x_1, x_2 \geq 0,\ x_3\ is\ unrestricted\ in\ sign$

13. [A] / [CO5]
Solve the transportation problem using Vogel's Approximation Method (VAM).
Identify whether degeneracy occurs and explain how it can be resolved.

	D1	D2	D3	D4	Supply
S1	8	6	10	9	20
S2	9	12	13	7	30
S3	14	9	16	5	25
Demand	10	15	20	30	

14. The profit per unit for transporting goods from 3 sources to 3 destinations is: [A] / [CO5]

	D1	D2	D3	Supply
S1	18	14	16	40
S2	12	20	15	50
S3	16	18	22	60
Demand	50	40	60	

a) Convert the profit matrix into an equivalent cost matrix.

b) Obtain an initial basic feasible solution.

c) Test for optimality and compute maximum profit.

15. Solve the following assignment matrix to achieve an optimal solution. List the specific machine assigned to each job: [U] / [CO6]

	Machine X	Machine Y	Machine Z
Job A	16	12	14
Job B	22	18	25
Job C	15	14	18

16. State the two theorems used in the Assignment Problem to find an optimal solution. [K] / [CO6]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. A firm manufactures three products A, B and C. Time to manufacture product A is twice that for B and thrice that for C and if the entire labour is engaged in making product A, 1,600 units of this product can be produced. These products are to be produced in the ratio 3: 4: 5. There is demand for at least 300, 250 and 200 units of products A, B and C and the profit earned per unit is ₹90,₹40 and ₹30 respectively. [U] / [CO1]

Raw material	Requirement per unit of product (kg)			Total availability(kg)
	A	B	C	
P	6	5	2	5,000
Q	4	7	3	6,000

(a) Formulate the linear programming problem..

(b) Show how ratio constraints are converted into linear equations.

18. Explain basic solution, feasible solution, basic feasible solution, degenerate and non-degenerate basic feasible solution, and discuss their importance in linear programming. [U] / [CO3]

19. Use Big M method, find minimum two iterations of given linear programming problem. [A] / [CO3]

$$\begin{aligned} \text{Maximize} \quad & Z = 3x_1 - x_2 \\ \text{subject to,} \quad & 2x_1 + x_2 \geq 2 \\ & x_1 + 3x_2 \leq 3 \\ & x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$

20. Obtain an optimum solution using MODI method for the following transportation problem [E] / [CO5]

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	34

21. a)Explain the steps in Hungarian method for solving assignment problem. [U] / [CO6]
b)Draw the flow chart for Hungarian method.

22. Explain the meaning of an Assignment Problem and write its full mathematical model. [U] / [CO6]

[10x3 = 30]